PCA: From Reconstruction Error to Principal Components

Work with a partner. Show your calculations and explain your reasoning. Use sample variance and covariance (divide by $n-1$). Assume all matrices below have observations in rows.

Part I: Revisiting the slightly scattered dataset

(1) Reconstructing from one direction.

Our centered data matrix is

$$X_{\text{new}} = \begin{bmatrix} 2 & 2 \\ 1 & 1.5 \\ -1 & -1.5 \\ -2 & -2 \end{bmatrix}, \qquad \mathbf{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

(a) For the observation $\mathbf{x} = (-1, -1.5)^T$, compute its scalar projection

$$z_2 = \mathbf{x}^T \mathbf{v}_1$$

and reconstruction

$$\hat{\mathbf{x}}_2 = z_2 \mathbf{v}_1.$$

(b) Calculate the squared reconstruction error

$$\|\mathbf{x} - \hat{\mathbf{x}}_1\|^2.$$

What part of the observation was lost?

(2) Comparing two possible directions.

(a) Repeat the projection and reconstruction for $\mathbf{x} = (-1, -1.5)^T$ using $\mathbf{v}_2$. Calculate the squared reconstruction error.

(b) Complete the table for all four observations. The columns are scalar coordinates, not reconstructed two-dimensional points.

Observation $\mathbf{x}$	$z_1 = \mathbf{x}^T\mathbf{v}_1$	$z_2 = \mathbf{x}^T\mathbf{v}_2$
$(2, 2)$	$2\sqrt{2}$	0
$(1, 1.5)$		
$(-1, -1.5)$		
$(-2, -2)$		

(c) Calculate the sample variance of each set of projected coordinates. Since the data is centered, use

$$\mathrm{Var}(\mathbf{z}) = \frac{1}{n-1}\sum_{i=1}^{n} z_i^2.$$

(d) Compute the *total* squared reconstruction error over all four observations for each direction. You may use

$$\sum_i \|\mathbf{x}_i\|^2 - \sum_i z_i^2$$

to simplify the calculation.

(e) Which direction gives less reconstruction error? Which has greater variance? Why is this connection useful for PCA?

(3) Where does the covariance matrix enter?

(a) Verify that the columns of X_{new} have mean zero. Calculate the sample variance of each feature and their sample covariance.

(b) Assemble the covariance matrix C_{new} from those three quantities.

Then verify it using

$$C_{\text{new}} = \frac{X_{\text{new}}^T X_{\text{new}}}{n-1}.$$

(c) We can express the variance of the projected observations in matrix notation:

$$\text{Var}(X_{\text{new}}\mathbf{v}) = \frac{(X_{\text{new}}\mathbf{v})^T(X_{\text{new}}\mathbf{v})}{n-1} = \mathbf{v}^T C_{\text{new}}\mathbf{v}.$$

Explain why

$$(X_{\text{new}}\mathbf{v})^T(X_{\text{new}}\mathbf{v})$$

equals the sum of squared projection coordinates.

Show the transpose step that produces

$$\mathbf{v}^T X_{\text{new}}^T X_{\text{new}}\mathbf{v}.$$

(d) Check whether $\mathbf{v}_1$ is *exactly* an eigenvector of C_{new}.

Does $C_{\text{new}}\mathbf{v}_1$ equal a scalar multiple of $\mathbf{v}_1$?

Explain why a visually plausible direction need not be the exact first principal component.

Part II: Perform PCA from start to finish

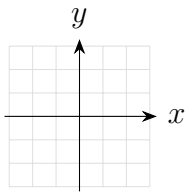
(4) A new dataset.

A dataset contains four observations and two numerical features:

$$Y = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

(a) Calculate the mean of each feature, then subtract each column mean to obtain the centered matrix Y_c.

(b) Draw the four observations. Based only on the plot, draw a direction that might be a good one-dimensional summary.



(c) Compute the covariance matrix

$$C = \frac{Y_c^T Y_c}{n-1}.$$

Show your matrix multiplication.

(5) Find the principal directions.

(a) Solve

$$\det(C - \lambda I) = 0$$

for both eigenvalues.

Show the characteristic equation and your algebra.

(b) For each eigenvalue, solve

$$(C - \lambda I)\mathbf{v} = 0$$

and normalize the eigenvector.

Check that the two unit eigenvectors are perpendicular.

(6) Reduce the dimension and reconstruct.

Use your eigenpairs from Problem 5.

(a) Which unit eigenvector is the first principal component direction?

Explain your choice using the eigenvalues.

(b) Let V_1 contain that unit eigenvector as its only column.

Compute the one-dimensional principal component scores

$$Z = Y_c V_1$$

for all four observations.

(c) Reconstruct the original (uncentered) observations using only the first component.

First compute

$$\hat{Y}_c = Z V_1^T,$$

then add the original feature means back to every row to obtain $\hat{Y}$.

Show at least one row calculation and give $\hat{Y}$.

(d) Compute the total squared reconstruction error

$$\sum_i \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|^2$$

on the original (uncentered) observations.

Compare it with reconstruction using only the *second* principal component.

(7) Interpret PCA.

(a) What fraction of the total variance is captured by the first principal component?

What fraction is discarded?

(b) In one or two sentences, explain why the direction with the largest covariance eigenvalue gives the better one-dimensional reconstruction.

Exit question: Write the steps of PCA starting from a dataset Y and ending with the lower-dimensional scores Z.